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Thermomechanical properties of carbon fibres at high temperatures (up to 2000 °C)

Cédric Sauder, Jacques Lamon*, René Pailler

Laboratoire des Composites Thermostructuraux, UMR 5801 (CNRS-SNECMA-CEA-Université Bordeaux I), 3 allée de La Boétie, 33600 Pessac, France

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Abstract

A high-temperature fibre-testing apparatus has been designed. It is dedicated to the determination of various properties at very high temperatures, including electrical conductivity, Young's modulus, thermal expansion coefficient, strength. Test temperatures as high as 3000 °C can be applied to carbon fibres. Two types of carbon fibres (a PAN-based and a Rayon-based fibre) have been investigated at temperatures up to 2000 °C. The measured properties are discussed with respect to microstructural features. © 2002 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Carbon fibres represent a large part of the materials used in aeronautics and aerospace structures, and in the reinforcement of composites (C/C brakes, etc.) They display a very wide range of thermal, electric and mechanical properties: elastic moduli vary between 30 and 900 GPa, strength can be as high as 6 GPa.

Reliable and accurate data for the mechanical properties of carbon fibres and matrices are a prerequisite for the development of valid models of thermomechanical behaviour of C/C composites that are based on micromechanics and constituent properties. Very few data are available in the literature for the properties of carbon fibres and matrices at high temperatures [1,2].

Mechanical tests on fibres with diameter $\leqslant 7$ μm or microcomposites with a diameter ≈ 30 μm are very difficult. Difficulties are enhanced under high-temperature conditions.

The present paper proposes an experimental approach for measuring the thermomechanical properties of carbon fibres at temperatures up to 3000 °C. Temperature dependence of properties observed on two types of carbon fibres are discussed with respect to microstructure.

* Corresponding author.

E-mail addresses: sauder@lcts.u-bordeaux.fr (C. Sauder), lamon@lcts.u-bordeaux.fr (J. Lamon), pailer@lcts.u-bordeaux.fr (R. Pailler).

2. Experimental procedure

The available methods for fibre testing at high temperatures exhibit various limitations. Two methods are essentially used: in the hot-grip technique, the whole of the fibre is exposed to a uniform temperature, whereas in the cold grip technique only a short part of the fibre is at the test temperature. Young's modulus and strength data (tensile strength, σ_R, and statistical parameters) can be measured by using the hot-grip method. The main limitation of this method lies in gripping. The use of a ceramic glue excludes tests at temperatures above 1500 °C.

The cold-grip method [3] exhibits several short comings. As a result of the presence of a large thermal gradient along fibre, determination of true strain is uncertain. Moreover, the method of correcting fibre elongation measurements is complex and inaccurate, so that results are obtained with a large uncertainty. Furthermore, since long specimens are required (longer than 10 cm), experimental difficulties are enhanced.

2.1. Experimental device

It is for all of the above reasons that a new apparatus for tensile tests at high temperatures was designed. Novelty can be found in the technique of specimen heating, which takes advantage of the electric conductivity of carbon fibres. An electric current is applied to the fibre, which allows temperature as high as 3000 °C to be

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reached, and a uniform temperature along the fibre to be generated. Temperature in the fibre is measured by using a bichromatic pyrometer in the 1000–3000 °C range. In practice, temperatures in the 1200–3000 °C range only are detected on fibres having a 7 μm diameter. Since tests are performed under secondary vacuum (residual pressure $<10^{-3}$ Pa) there is a relationship between fibre temperature and the electric power supplied as shown in Fig. 1. As a consequence, temperature of fibre can be easily derived from the electric power.

Displacement of grips are measured by a sensor whose sensitivity is smaller than 0.1 μm. Fibre strain is derived from grip displacement by using a compliance calibration technique that allows deformation of load strain to be taken into account [4]. A technique of direct measurement of fibre elongation is under development. It is based upon optical extensometry and image analysis.

A schematic diagram of the high temperature fibre testing apparatus is shown in Fig. 2. Alignment of grips is improved using internal and external devices. The load cells are located within the vacuum chamber and all the experimental data are recorded by a computer.

2.2. Materials and specimen preparation

Two different carbon fibres were investigated: a rayon-base and a polyacrylonitrile-based fibre. Fibres were tested either as-received or after treatment at 1600 or 2200 °C. Fibres were fixed to graphite grips using a cement (C-34, UCAR). Batches of 20 specimens were prepared, except for the tests at 1200, 1400 and 1800 °C on the rayon-based fibres.

Fibre diameter was measured before tests using laser diffractometry. The fibre was mounted on the testing apparatus. Fibre section was assumed to be circular. This is true for the PAN-based fibre.

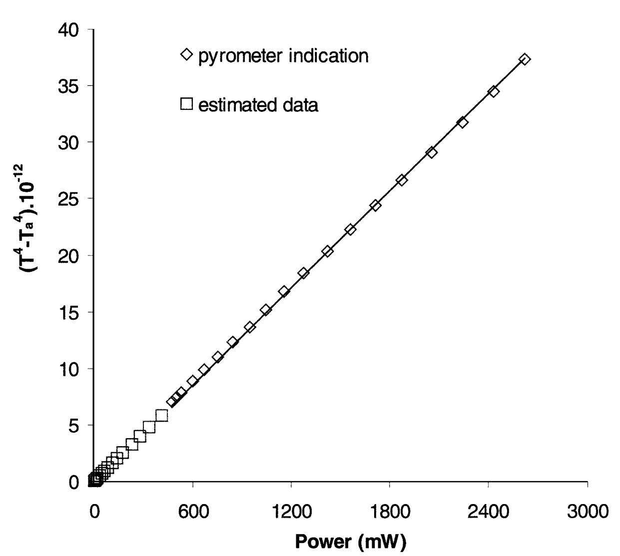

Fig. 1. Correlation between electrical power applied to the fibre and the temperature measured by using a pyrometer.

2.3. Experimental procedure

2.3.1. Measurement of electrical conductivity

The calibration curve established for temperature determination (Fig. 1) allows determination of electrical conductivity (σ) at various temperatures:

$$\sigma(\Omega^{-1}\mathrm{m}^{-1}) = \frac{1}{\rho} = \frac{L}{R.S}$$

Where $\rho = R.S/L$ (resistivity), where R is the resistance ($R = U/I$, U = tension, I = intensity), S = sectional area and L = fibre length.

2.3.2. Measurement of thermal expansion

Very few data on thermal expansion of carbon fibres are available at high temperatures [2,5,6]. The technique used here is very simple. A very low stress is applied onto the fibre (a few MPa). This stress is maintained constant during fibre heating through control of grip displacement. The displacement which is applied balances the longitudinal expansion of fibre induced by heating. Such thermal expansion measurements are very easy and very accurate under computer controlled testing conditions.

2.3.3. Mechanical behavior

Tensile tests were performed on those PAN-based and rayon-based fibres which had been treated at 2200 °C. The range of test temperatures could thus be increased from 1000 to 2200 °C, since mechanical properties of carbon fibres depend on the heating rate when the test temperature is higher than the treatment temperature. Gauge length was 50 mm. The Young's modulus was

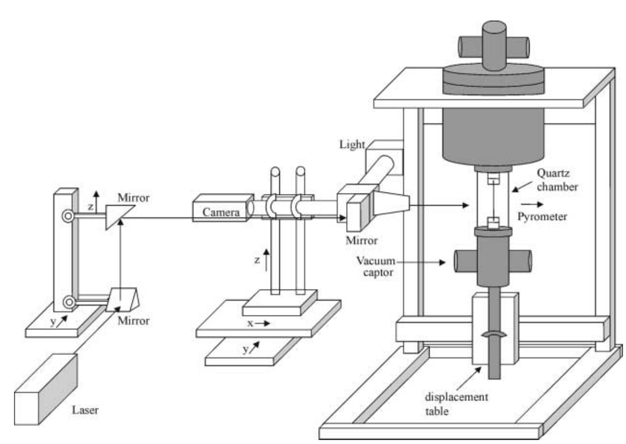

Fig. 2. Schematic diagram of the high temperature fibre testing apparatus.

determined under a 0.2% strain. The statistical parameters (Weibull modulus m and scale factor σ_0) were estimated from the statistical distributions of strength data using a linear regression technique [7].

3. Results and discussion

3.1. Electrical conductivity

Electrical conductivity measured at various temperatures on the PAN-based and rayon-based fibres, that had been previously treated at 2200 °C, are plotted in Fig. 3a and b. The temperature dependence of electrical conductivity is typical for these fibres, since it can be noticed that electrical conductivity increases with temperature. This trend characterizes a semiconductor. Although data at low temperatures were not determined, it can be considered that this trend is pertinent to an extrinsic semiconductor. Indeed, at low temperatures, conduction is governed by impurities (extrinsic conductivity), whereas at high temperatures the thermal energy is sufficient to induce electronic transitions (strip of valence B.V.→strip of conduction B.C.). This is the domain of intrinsic conductivity for this type of carbon microstructure. The energy gap ($\Delta E = E_c - E_v$) can be determined by using the following equation:

$$\sigma = C\exp(-\Delta E/2kT)$$

$\Delta E = 0.12$ and 0.18ev were obtained respectively for the PAN-based and the rayon based fibres. These values are quite small.

3.2. Longitudinal thermal expansion

In order to check the validity of measurements, the technique was applied to a tungsten fibre having a 18 µm diameter. Fig. 4 shows that the measured thermal expansions coincide with those values derived from the coefficient of thermal expansion available in the literature (SETARAM data).

Thermal expansions at various temperatures are plotted in Figs. 5 and 6. It can be noticed that the investigated fibres exhibit different behaviors. Expansion of the PAN-based fibre is much smaller than that of the rayon-based fibre. At temperatures close to ambient temperature, the PAN-based fibre contracts. This behavior is comparable to that displayed by graphite single crystal. It is characterized by a negative coefficient of thermal expansion at 20 °C. This behavior results from the fibre microstructure shown in Fig. 7 [8], which is not isotropic but instead involves basal structural units (BSU) with a preferred orientation with respect to the fibre axis. This organized microstructure contrasts with the one of rayon-based fibre. Fig. 6 also shows that treatment influences the fibre response.

This phenomenon can be attributed to successive changes in the fibre microstructure. As the treatment temperature increases, the BSU tend to be oriented parallel to the fibres axis, which would increase the coherent domains in X-ray diffraction (XRD). On the contrary, the behavior and the microstructure of the rayon based fibres do not change when the treatment temperature is increased.

Expansion of graphite single crystal is shown in Fig. 8 [9]. It can be noticed that for both fibres, longitudinal thermal expansion results from a combination

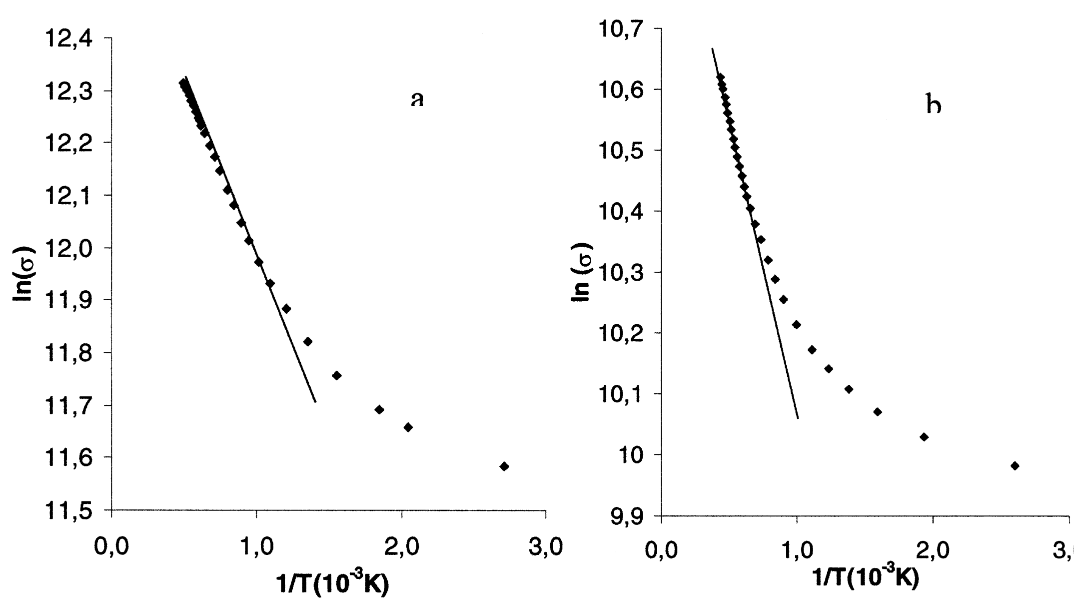

Fig. 3. Electrical resistivity vs temperature for a PAN-based (a) and a rayon based (b) fibre.

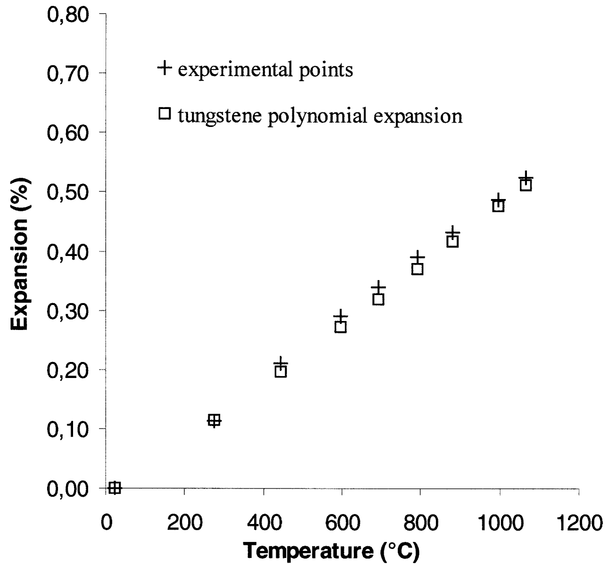

Fig. 4. Longitudinal thermal expansion at various temperature for a tungsten fibre (d= 18 μm).

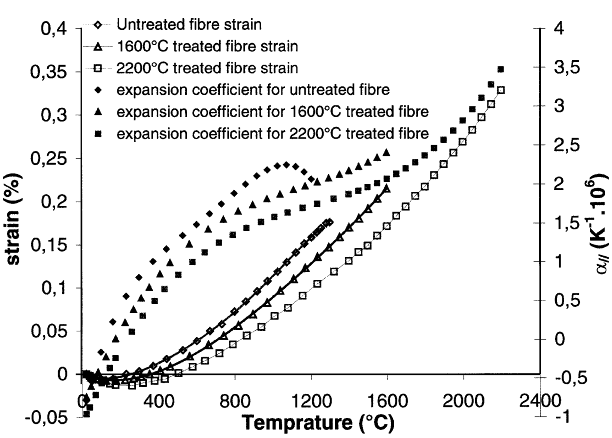

Fig. 5. Longitudinal thermal expansion at various temperatures for a PAN-based fibre.

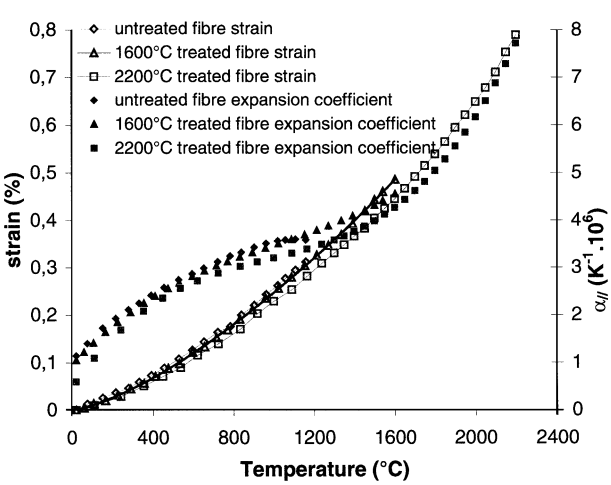

Fig. 6. Longitudinal thermal expansion at various temperatures for a rayon-based fibre.

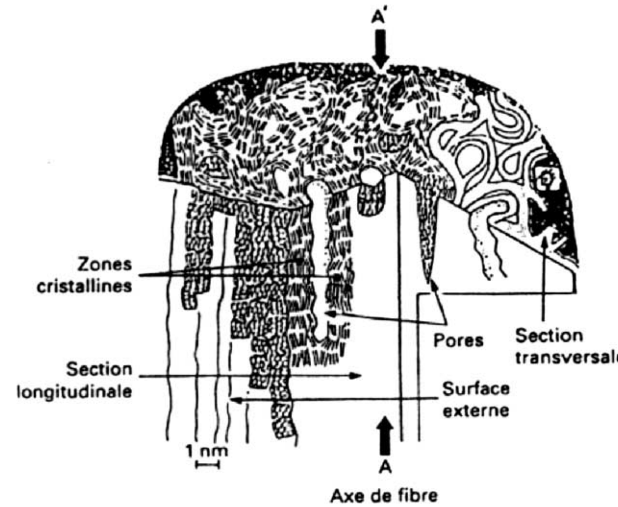

Fig. 7. Structure of an ex-PAN fibre [8].

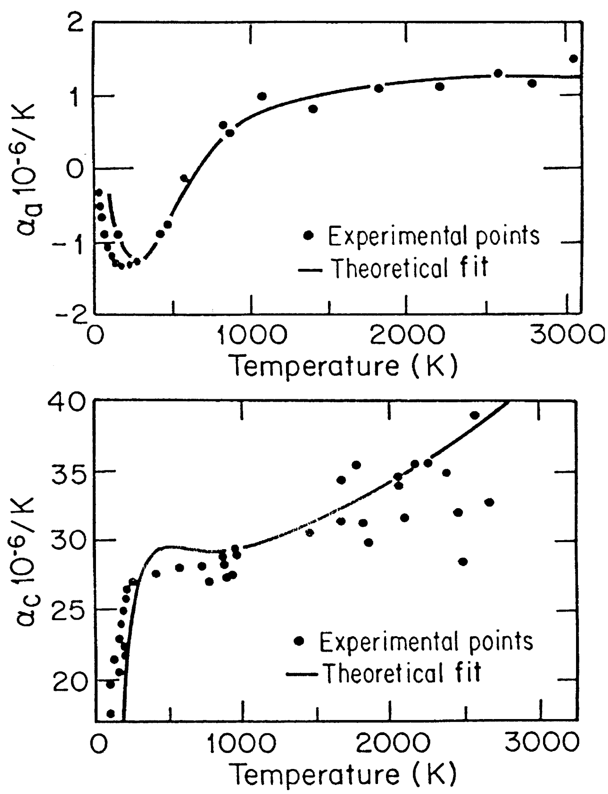

Fig. 8. Bulk graphite thermal expansion [9].

of thermal expansions of single crystal in the transverse and in the longitudinal directions. Indeed, if orientation of BSU was parallel to fibre axis, the expansion of fibre would be close to that indicated by α_a. However, at temperatures above 1000 °C the coefficient of thermal expansion still increases, which suggests that BSU are not perfectly oriented parallel to fibre axis and that α_c governs thermal expansion in the longitudinal direction. This phenomenon is particularly significant at temperatures above 1000 °C. At these temperature α_c increases

Table 1
Mechanical properties at various temperatures

Temperature °C	PAN-based fibre					Rayon-based fibre				
	E/E_0 (%)	σ_R (MPa)	ε_R (%)	m	σ_0[a] (MPa)	E/E_0 (%)	σ_R (MPa)	ε_R (%)	m	σ_0*(MPa)
24	100	2107 (405)[b]	0.71 (0.13)	5.1	660	1	719 (157)	2.16 (0.45)	5.1	230
1000	98.5 (0.4)	2292 (410)	0.80 (0.14)	6.5	950	96.6 (0.4)	789 (103)	2.39 (0.31)	8.3	390
1200	97.7 (0.5)	2401 (540)	0.86 (0.18)	5.2	890	94.6 (0.4)	756 (118)	2.32 (0.38)	ND	ND
1400	95.7 (0.8)	2386 (507)	0.85 (0.18)	5.4	810	89.5 (0.6)	825 (116)	2.68 (0.36)	ND	ND
1600	91.6 (1.6)	2359 (520)	0.91 (0.23)	5.4	800	83.2 (1.0)	821 (114)	5.41 (1.75)	7.9	390
1800	81.8 (1.7)	2857 (611)	1.53 (0.44)	5.6	1020	69.4 (1.0)	605 (72)	12 (6.7)	ND	ND
2000	69.5 (1.8)	2348 (443)	3.71 (1.84)	5.9	850	52.7 (1.7)	371 (39)	12.7 (6,1)	10	200

[a] Vo = 1 mm^3.
[b] ( ) Standard deviation.

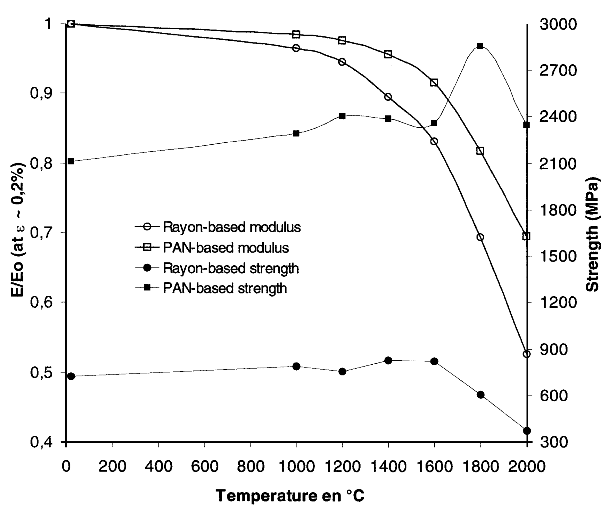

Fig. 9. Young's modulus and tensile strength at various temperatures for a PAN-based fibre and a rayon-based fibre.

strongly whereas α_a becomes constant. Thus it appears that longitudinal thermal expansion provides indications on the orientation of BSU and the degree of organization of fibre microstructure.

3.3. Mechanical behavior

Mechanical properties measured at various temperatures are summarized in Table 1. The elastic modulus of both fibres exhibits the same trend (Fig. 9). The elastic modulus is quite constant at temperatures $\leqslant 1000$ °C, then it decreases steeply at temperatures >1000 °C. This phenomenon is more significant for the rayon-based fibre. A similar phenomenon has been previously reported for a Thornel 50 fibre [10]. It may be attributed to an increase in anharmonic vibrations that would lead to plastic deformations and a reduced linear elastic behavior. This phenomenon becomes less significant as BSU are oriented parallel to fibre axis.

A different trend is observed for the strength σ_R (Fig. 9). For both fibres σ_R increases slightly when the temperature increases to 1600 °C. This phenomenon may result from a reduction in flaw severity associated with internal stress relaxation.

At temperatures >1600 °C, fibres exhibit a different behavior. For the rayon-based fibre, the strength decrease is large whereas the strength scatter narrows (Table 1). For the PAN-based fibre, σ_R reaches a maximum (2850 MPa) at 1800 °C and then decreases to a value close to that measured at room temperature. The origin of this phenomenon has not been elucidated yet. This will require specific analyses that fall out of the scope of this paper.

Finally, the statistical parameters (Weibull modulus m and scale factor σ_o) indicate that flaw population in the PAN-based fibre is not influenced by temperature. On the contrary, the flaw population seems to become more homogeneous in the rayon-based fibre as temperature increases, since m increased from 5 to 10 when the temperature was increased from 24 to 2000 °C.

4. Conclusions

The high temperature fibre testing apparatus that was developed allowed determination of a large variety of carbon fibre properties at temperatures up to 2000 °C: electrical conductivity, longitudinal thermal expansion, Young's modulus and strength. Tests at temperatures as high as 3000 °C can be performed.

Temperature dependence of thermal expansion and mechanical properties reflected the preponderant contribution of microstructure of fibres to their properties ($\alpha_{//}$, E, σ_R, m, σ_0). In those PAN-based fibres, "BSU" tend to be oriented parallel to fibre axis. These fibres are less sensitive to temperature than the rayon-based fibres. Besides, PAN-based fibres contract at temperatures close to ambient temperature. This behavior is close to that displayed by graphite single crystal.

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